# CATH PLAN · Case 3

59M, 102 kg · scheduled 10:00 · diagnostic cath ± PCI (exertional chest pressure) · pathway: indeterminate, pre-cath plan

**Synthetic teaching content. Not clinical advice. Not a medical device. DRAFT for the chair to rule on.**

**PRE-CATH PLAN: angiography has not been done. The band is a label from a proposed points rule, not a probability, and the packet lists equipment for both branches.**

**Thresholds: PROPOSED values, not yet verified by the chair. Template logic (likelihood weights and bands, prep rules, Imaging triggers, Heart Team factors, stage assignment, Labs' eGFR cutoff) is also unreviewed.**

| 7 | n/a | higher | 1 |
|---|---|---|---|
| flags from 4 agents | contrast ratio (Laskey 2007 discriminator 3.7, not a limit) | points band (9 pts, proposed rule) | devices out of stock |

Contrast ratio = planned contrast (mL) ÷ eGFR. The published ratios use creatinine clearance, not eGFR (docs/EVIDENCE.md E1, E2): risk was flat below 2 and rose above 3 (Gurm 2011) and 3.7 was a discriminating cutoff (Laskey 2007). The plan shows both the eGFR ratio and, when age, sex, weight and creatinine are in the packet, a Cockcroft-Gault ratio, and applies the cutoffs to the higher. No guideline sets a limit; the chair confirms the basis.

## Pre-cath findings (CCTA, stress, symptoms)

| CCTA lesion | Vessel | Stenosis | Calcium |
|---|---|---|---|
| L1 | mid-LAD | 75% | calcium none |
| L2 | prox-LCx | 55% | calcium moderate |

Stress test: positive stress echo, anterior wall · EF: 55% · Symptoms: typical angina

| Points driver (proposed points rule) | Points |
|---|---|
| CCTA 75%, stenosis of 70% or more (the CAD-RADS 4A/4B/5 range; CAD-RADS categories are per patient) (L1 mid-LAD; E5) | +2 |
| CCTA 55%, intermediate stenosis (L2 prox-LCx; E6) | +1 |
| two or more CCTA lesions | +1 |
| stress test positive (stress echo, anterior wall) | +2 |
| typical angina | +1 |
| the positive stress result (anterior wall) is in the territory of the CCTA lesion | +2 |
| **Total · band: higher** | **9** |

A label from a points rule, not a probability. The chair decides whether to go on to PCI.

## Lesion map (Imaging)

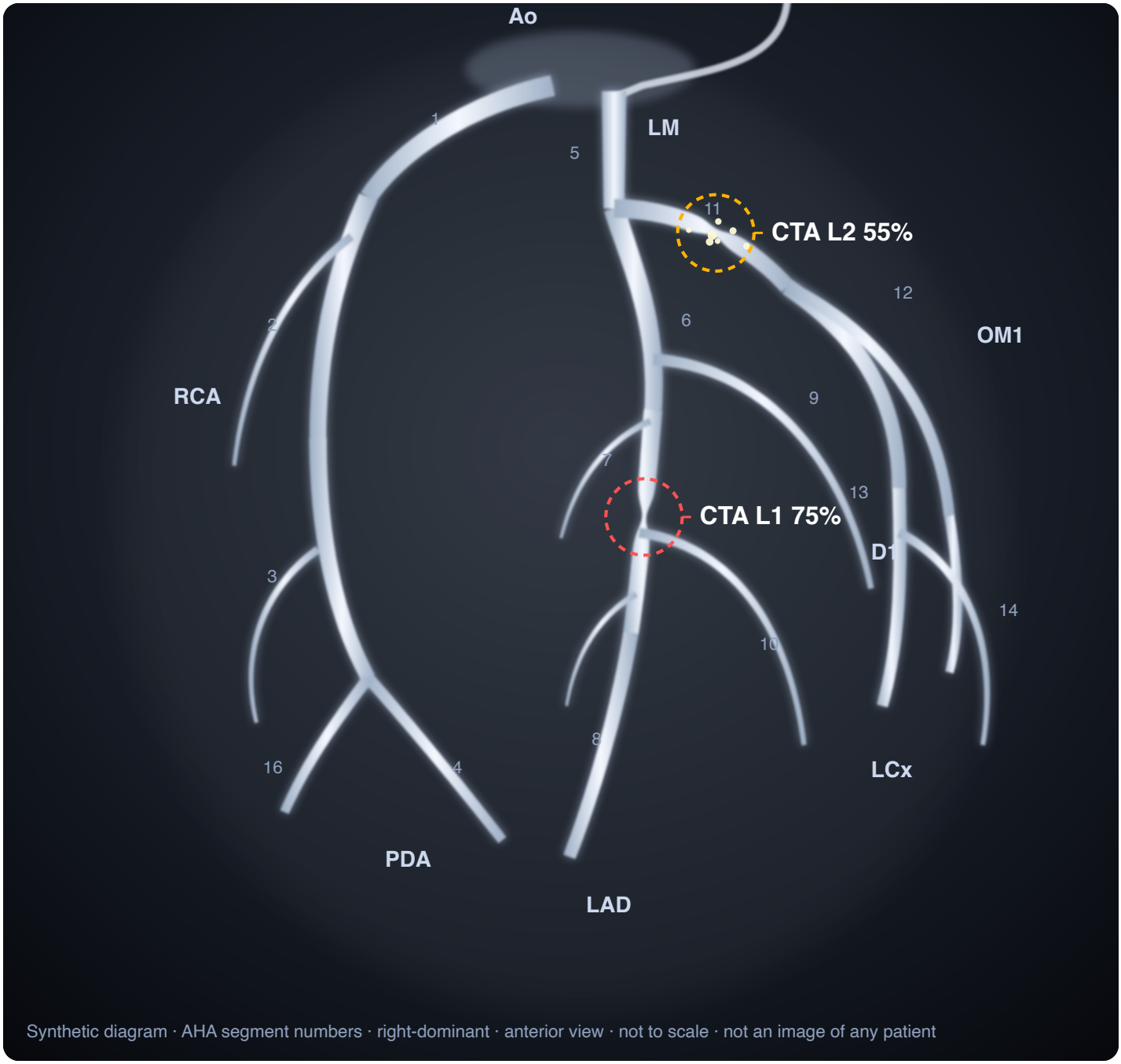
| Lesion | Vessel | Stenosis | TIMI | Calcium | Flag |
|---|---|---|---|---|---|
| Angiography not yet performed: this is a pre-cath plan | | | | | |

Angio findings: not assessed (angiography not yet performed). · Cath report: not in packet

## Rules that fired

- Points band (proposed rule, 9 points): higher. The band word only labels a points range; it is not a probability and not a clinical pre-test probability. Chair to decide; the packet lists equipment for both branches below.

## Pre-cath CCTA findings (synthetic, not yet confirmed by angiography)




**L1 · mid-LAD** (AHA segment 7)
CCTA estimate 75% · calcium none · not yet confirmed by angiography

**L2 · prox-LCx** (AHA segment 11)
CCTA estimate 55% · calcium moderate · not yet confirmed by angiography

Drawn from the angio lines in the packet by tools/angio.py: right-dominant tree, anterior view, AHA segment numbers, lumen narrowed to the stated percentage, calcium shown as bright specks, distal filling dimmed when TIMI flow is below 3. Solid red ring = stenosis of 70% or more; dashed amber ring = under 70%, or a CCTA estimate not yet confirmed. A Medina 1,1,1 flag is drawn as a narrowing of about half the proximal LAD and LCx. A teaching diagram, not an image of any patient and not to scale.

## Equipment manifest

| Branch A · stays diagnostic | |
|---|---|
| **Item (from the packet)** | **Stock** |
| 5F diagnostic catheters | 8 on hand |

| Branch B · proceeds to PCI | |
|---|---|
| **Item (from the packet)** | **Stock** |
| 6F guide | 4 on hand |
| NC balloon 3.0 mm | 2 on hand |
| DES 3.0x18 mm | **OUT: 0 on hand · no resupply scheduled** |
| standard PCI tray | 4 on hand |

## Prep check (devices against findings)

| | |
|---|---|
| **Scope** | The packet lists Branch A and Branch B (band higher, proposed rule) |
| **Stock** | **OUT: DES 3.0x18 mm** |

| On the shelf, by class (from the STOCK line) | |
|---|---|
| Guide catheters | 6F guide 4, 7F guide 2 |
| Diagnostic catheters | 5F diagnostic catheters 8 |
| Wires | support wire 4 |
| Balloons | NC balloon 3.0 mm 2, NC balloon 3.5 mm 2 |
| Stents (DES) | DES 3.5x24 mm 3, DES 3.5x28 mm 2, DES 3.0x18 mm 0 (no resupply scheduled) |
| Intravascular imaging | IVUS catheter 1 |
| Support devices | Impella CP 0 (no resupply scheduled) |

**Questions for the chair** (the plan asks, it does not choose a device):

- 2 lesions on CCTA and 1 stent named in the packet: chair to confirm what to stage.
- Calcium moderate on L2 prox-LCx (CCTA): the packet names no calcium-modification device or intravascular imaging catheter. Chair to decide whether any is needed.
- No wire is named in the packet's device lines: chair to confirm.

## Flags from the agents

**Imaging**
- CCTA 75% · L1 mid-LAD · ref: CAD-RADS 2.0
- CCTA 55% · L2 prox-LCx · ref: ESC 2024 intermediate CCTA
- stress positive · anterior wall

**Labs**
- Hgb 9.8 vs ref 13.5-17.5
- INR 2.6 vs ref 0.9-1.1

**Meds**
- warfarin last dose 14 hours before the 06:30 report · diagnostic cath ± PCI

**Equipment**
- DES 3.0x18 mm 0 on hand · DEVICES IF PCI lists "DES 3.0x18 mm" · STOCK: "no resupply scheduled"

Relay: Imaging → Equipment, Labs → Meds.

## Summary and approvals

| | |
|---|---|
| **Ruling recorded** | none yet |
| **Flags** | Imaging 3 · Labs 2 · Meds 1 · Equipment 1 |
| **Rules that fired** | Points band (proposed rule, 9 points): higher. The band word only labels a points range; it is not a probability and not a clinical pre-test probability. Chair to decide; the packet lists equipment for both branches below. |

**For the chair to decide / check** (the agents only flag):

- The points rule is proposed: the chair verifies it.
- The packet lists equipment for both branches: confirm stock for the branch actually taken.
- Thresholds in thresholds.json are proposed: the chair verifies them before this plan is used.
- Template logic (likelihood weights and bands, prep rules, Imaging triggers, Heart Team factors, stage assignment, Labs' eGFR cutoff) is unreviewed.

**Approvals**

No approval decisions recorded yet.

For the chair's use: print and mark.

- ☐ Approve as drafted
- ☐ Chair changes the plan (record the change)
- ☐ Equipment issue noted

**Status**

This draft is not approved until the chair presses Allow once on the Switch card.

AWAITING CHAIR

Sources: packet (day2.md) · briefs from Imaging, Labs, Meds, Equipment · thresholds.json · evidence table docs/EVIDENCE.md (E1 to E16, mostly secondary summaries, chair verifies). Generated 2026-11-04 09:00 by plan.py (template, no model).

Provenance: template 7b83bc09dd · thresholds bea3b78f4a · plan.py fa5860885c · packet id n/a. "ref:" on a flag names a published criterion; it is not an instruction, and the criteria are unverified until the chair has checked them against the sources (docs/EVIDENCE.md).